
Lab 7: Árbol sumador

Goals

- Design sequential systems using Verilog HDL.
- Design arithmetic circuits with parallelism.

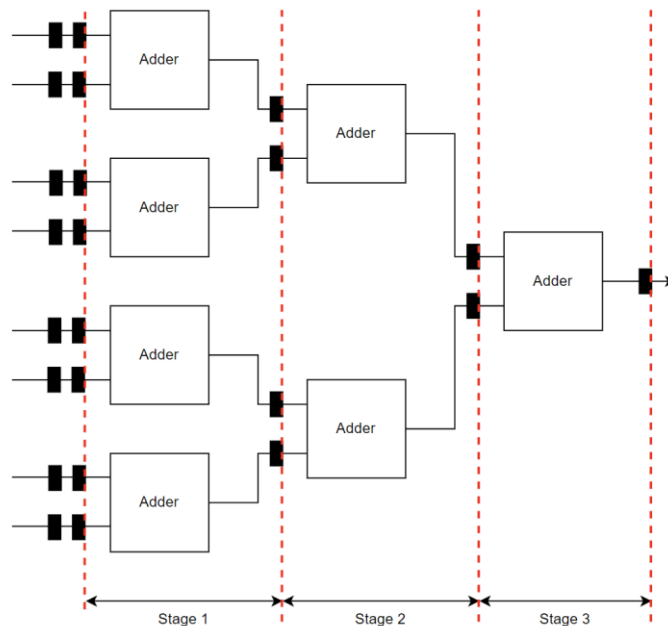
Introduction

A tree adder is an arithmetic circuit structure used to efficiently sum multiple operands, reducing computation time compared to a serial adder. It is organized in a hierarchy of adders that combine operands in parallel, thereby improving calculation speed.

Lab description

Design an adder tree with the following characteristics:

- The word size of the adders must be parameterizable.
- The adder tree must be capable of performing 8 additions in parallel.
- The adder tree must be segmented such that the output of each adder is connected to a register.



To verify the functionality of your design, use the following testbench code.

```

module adder_tree_tb;

parameter DATA_WIDTH = 32, TREE_INPUTS = 4;
bit clk;
logic [DATA_WIDTH-1:0] data_a [TREE_INPUTS];
logic [DATA_WIDTH-1:0] data_b [TREE_INPUTS];
logic [DATA_WIDTH-1:0] data_out;
localparam STAGES = $clog2(TREE_INPUTS)+1;

    adder_tree adder_tree_i (.clk(clk),.data_a(data_a),.data_b(data_b),.data_out(data_out));

always #5ns clk = ~clk;
int result;
reg [DATA_WIDTH-(STAGES+1):0] val_a;
reg [DATA_WIDTH-(STAGES+1):0] val_b;
initial begin
    repeat(100) begin
        @(negedge clk);
        result = 0;
        for (int n=0; n<TREE_INPUTS; n=n+1) begin
            val_a = $random;
            val_b = $random;
            data_a[n] = val_a;
            data_b[n] = val_b;
            result = result + val_a + val_b;
        end

        repeat (STAGES)
            @(negedge clk);
        $display("data_out=%0d",data_out);
        assert (data_out == result);
    end
    $finish;
end
endmodule

```

Additionally, for extra points, you shall submit a parameterized version of the adder_tree that can be configured for parameter N that is a multiple of 2. This would require changing the testbench too.

Submission

Implementation description

Generate a document providing a detailed description of the implementation. This should include an explanation of design decisions, challenges encountered, and how they were overcome. Include diagrams, schematics, tables, or equations used in the design development.

Video

Upload a short video to the Teams group, recording the operation of the implemented circuit on the FPGA and providing a brief description of its functionality.
